

Unit - I

Software Project Estimation and Decomposition



Software Project Estimation

- Effective software project estimation is important activity in software development project.
- Estimation in software engineering is a procedure that predicts the time and budget that required for completing a project.
- Project Estimation requires the use of complex tools & good mathematical as well as knowledge about planning.
- One of the main reasons software programs fail is our inability to accurately estimate software size.

➤ Responsible Persons:

- ☐ Software Manager
- ☐ Cognizant Engineers
- ☐ Software Estimators.

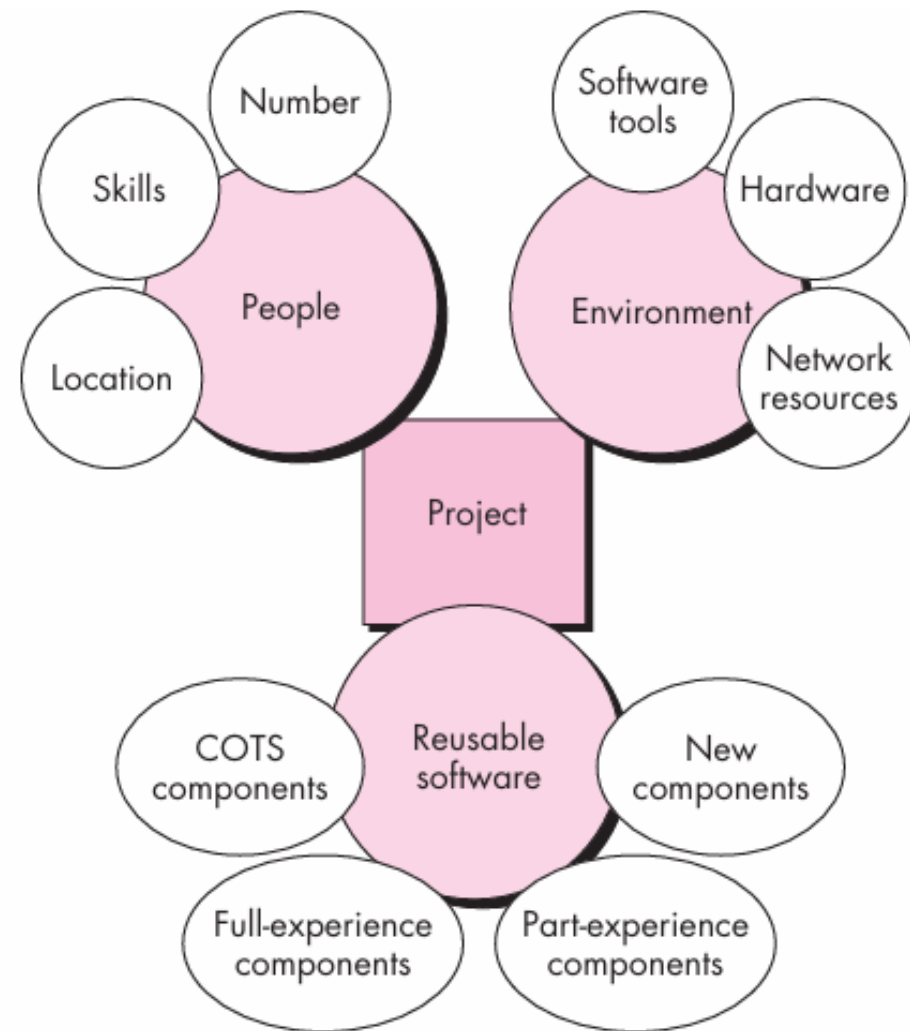


Factors affect on Project Estimation

1. **Cost:** The project will fail if you do not have sufficient funds to complete it. So, At early stage estimate project cost & ensure you have enough money to complete the work.
2. **Time:** Estimate both overall project duration & timing of individual tasks. It also enables you to manage client expectations for key deliverables.
3. **Size & Scope:** All the tasks that must be completed in order to deliver product on time. You can ensure that you have right materials & expertise on the project by estimating how much work is involved and exactly what tasks must be completed.
4. **Risk:** Estimating predicting risk, what events will occur during the project's life cycle and how serious they will be. Create risk management plans.
5. **Resource's:** Resource management ensures that you have all the resources you require & make best use of them. Like, tools, people, materials, hardware, software & other resources

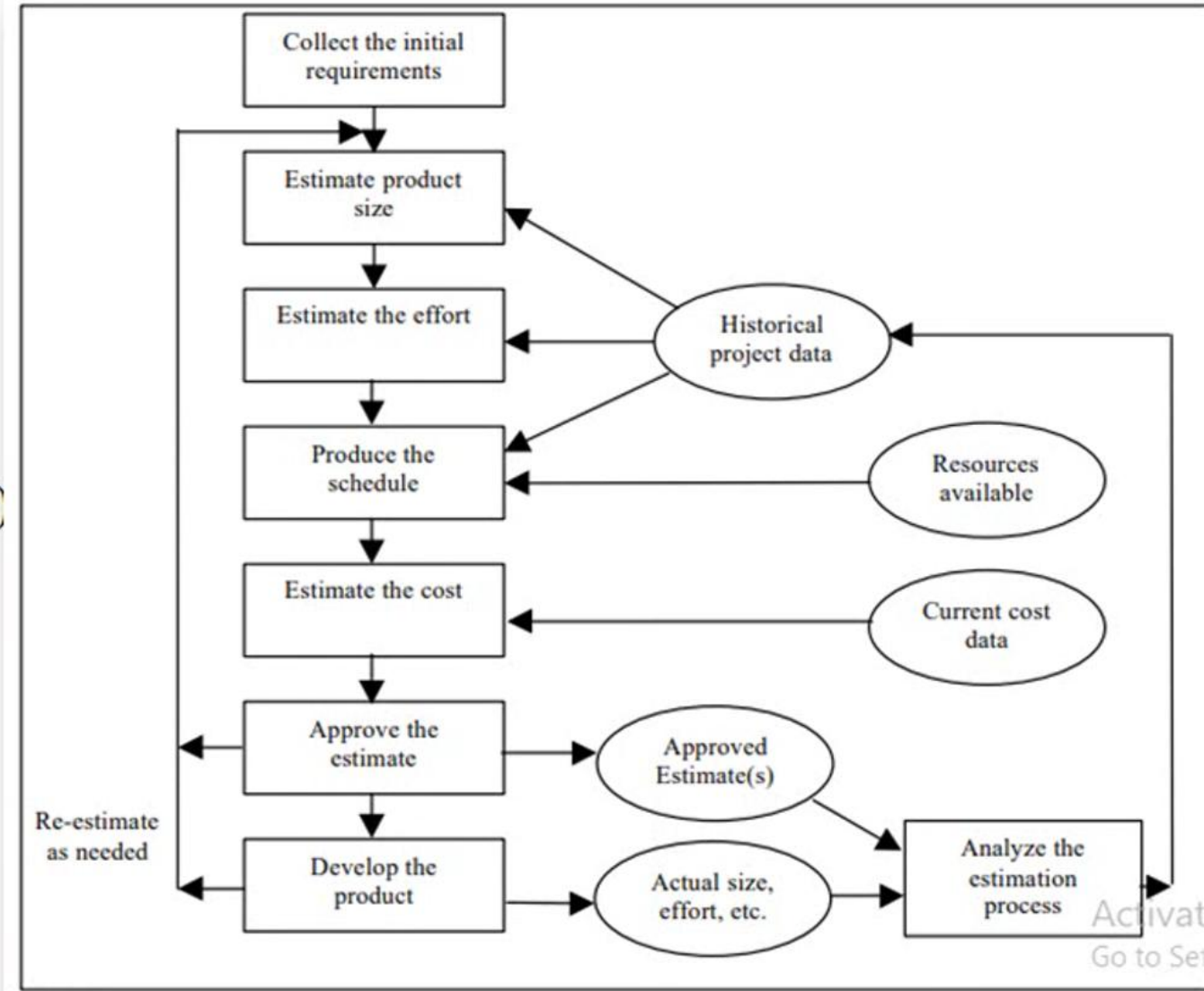
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Steps of Software Project Estimation

1. Estimate Project Size
2. Estimate Efforts (Person per month / hr)
3. Estimate Project Schedule (Month / Year)
4. Estimate Project Cost (Currency / Dollars)

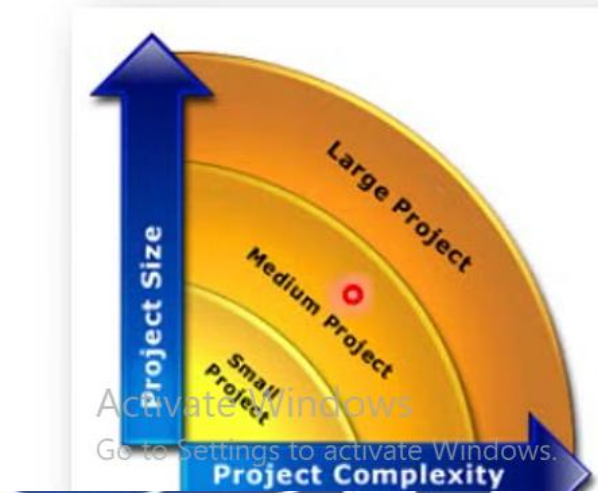


The Basic Project Estimation Process

Steps of Software Project Estimation

1. Estimate Product Size (LOC & FP):

- It is the very first step to make an effective estimation of the project.
- A Customer's requirements, SRS Document & System design document used for estimating the size of a software.
- Estimate project size can be through similar project developed in past. This is called estimation by Analogy.
- The system is divided into several subsystems depending on functionality & size of each subsystem is calculated.



Steps of Software Project Estimation



2. Estimate Project Efforts:

- The estimation of effort can be made from the organizational specifics of SDLC.
- Software development project involves Design, Coding, Testing, Writing, Reviewing documents, Implementing prototypes & Deliverable it decide overall project efforts.
- The project effort estimate requires you to identify and estimate & then sum up all the activities you must perform to build a product of the estimated size.

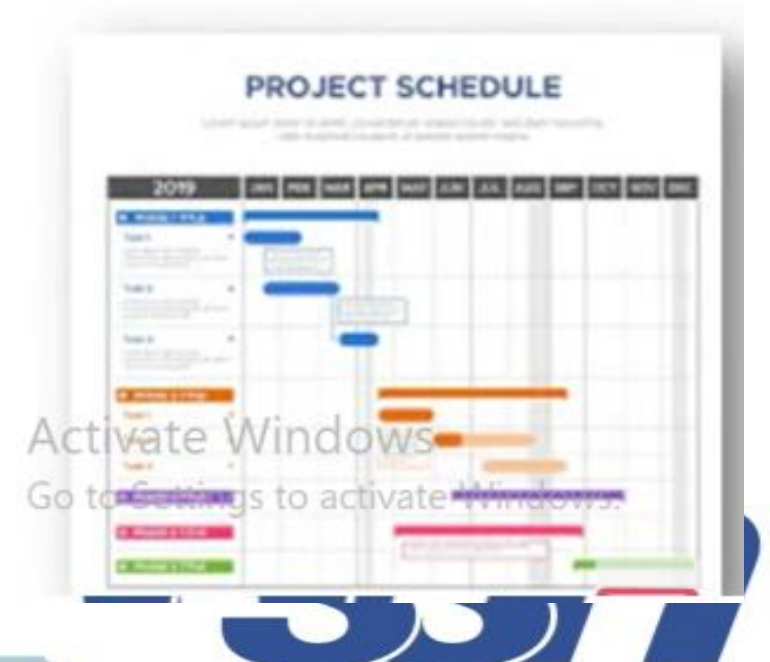
There are two main ways:

1. The best way to estimate effort is based on the organization's own historical data of development process. Suppose you have similar SDLC, Project size, Development methodology, Tools, Team with similar skills and experience for the new project.
2. If the project is in different nature which requires the organization to adopt different strategy or different models based on algorithmic approach. Ex. COCOMO Model.

Steps of Software Project Estimation

3. Estimate Project Schedule:

- It involves estimating number of people who will work on the project & what they will work on (the Work Breakdown Structure).
- Also when they will start working on project & when they will finish.
- Once you have this information, you need to lay it out into a calendar schedule.
- Efforts in man-month are translated to calendar months thumb rule is used.



Steps of Software Project Estimation

4. Estimate Project Cost:

- The cost of a project is derived not only from the estimates of person efforts and size.
- Other parameters such as purchase hardware, software, travel for meeting, telecommunication costs (long distance phone calls, video-conferences), training , office space etc.
- Exactly how you estimate total project cost will depend on how your organization allocates costs.
- The simplest labor cost can be obtained by multiplying the project's effort estimate (in hours) by a general labor rate (\$ per hour).
- A more accurate labor cost would result from using a specific labor rate for each staff position (e.g., Technical, QA, Project Management, Documentation, Support, etc.)



Decomposition Techniques

- **Software Project Estimation** is a form of problem solving (To estimate cost, time & efforts in software project.)
- **Decomposition Technique** is divide & conquer approach of Software Project Estimation.
- By decomposition a project into major functions like software engineering related activities, cost, schedule & efforts estimation can be performed in stepwise manner.

➤ Decomposition Techniques are:

1. Software Sizing
2. Problem based Estimation
3. Process based Estimation



1. Software Sizing

- Sizing represents the project planner's first major challenge.

➤ **The accuracy of a software project estimate is predicated on a number of things:**

1. The degree to which the planner has properly estimated the size of the product to be built.
2. The ability to translate the size estimate into human effort, calendar time and dollars.
3. The degree to which the project plan reflects the abilities of the software team.
4. The stability of product requirements & environment that supports software engineering effort.

➤ **Approaches of Software Sizing:**

1. “Fuzzy Logic” Sizing
2. Function Point Sizing
3. Standard Component Sizing
4. Change Sizing

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Approaches of Software Sizing

Putnam and Myers suggest four different approaches of Software Sizing Problem:

1. “Fuzzy Logic” Sizing:

- To apply this approach, the planner must identify the type of application.
- Although personal experience can be used, the planner should also have access to historical database of projects so that estimates can be compared to actual experience.

2. Function Point Sizing:

- The planner develops estimates of the information domain characteristics.

Approaches of Software Sizing

3. Standard Component Sizing:

- Standard components for an information system are subsystems, modules, screens, reports, interactive programs, batch programs, files, LOC and object-level instructions.
- Uses historical project data to determine the delivered size per standard component.

4. Change Sizing:

- This approach is used when a project encompasses the use of existing software that must be modified in some way as part of a project.
- The planner estimates reuse, adding code, changing code, deleting code of modifications that must be accomplished.
- Using an “effort ratio” for each type of change, the size of the change may be estimated.

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2. Problem based Estimation

- Lines of code and function points were described as measures from which productivity metrics can be computed. (LOC/pm & FP/pm)

- LOC and FP data are used in two ways during software project estimation:

1. As an estimation variable to "size" each element of the software.
2. As baseline metrics collected from past projects and used in conjunction with estimation variables to develop cost and effort projections.

➤ A three-point or expected value can then be computed

optimistic estimate - the shortest amount of time it is believed the task will take.

A most likely estimate - the most probable amount of time the task will take.

pessimistic estimate - the longest amount of time it is believed the task will take.

$$S = (s_{opt} + 4s_m + s_{pess})/6$$

– S = expected-value for the estimation variable (size)

– s_{opt} = optimistic value

– s_m = most likely value

– s_{pess} = pessimistic value

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Examples of LOC & FP

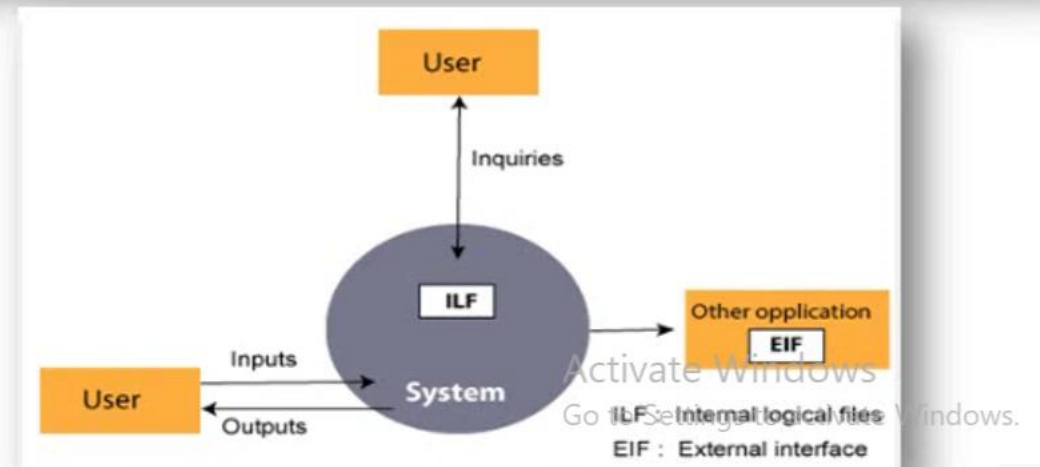
1. LOC (Lines of Code):

Project	LOC	Cost \$K	Efforts (Persons/ Month)	Documentation	Errors
A	10000	110	18	365	39
B	12000	115	20	370	45
C	15400	130	25	400	32

Function	Estimated LOC
User interface and control facilities (UICF)	2,300
Two-dimensional geometric analysis (2DGA)	5,300
Three-dimensional geometric analysis (3DGA)	6,800
Database management (DBM)	3,350
Computer graphics display facilities (CGDF)	4,950
Peripheral control function (PCF)	2,100
Design analysis modules (DAM)	8,400
<i>Estimated lines of code</i>	<i>33,200</i>

2. FP(Functional Points)

Information domain value	Opt.	Likely	Pess.	Est. count	Weight	FP count
Number of external inputs	20	24	30	24	4	97
Number of external outputs	12	15	22	16	5	78
Number of external inquiries	16	22	28	22	5	88
Number of internal logical files	4	4	5	4	10	42
Number of external interface files	2	2	3	2	7	15
Count total						320



Size Metrics : LOC

- Size-Oriented Metrics concentrates on the size of the program created.

1. LOC (Lines of Code):

- It is one of the earliest and simpler metrics for calculating the size of the computer program.
- It is generally used in calculating and comparing the productivity of programmers.
- LOC not count comment or blank line in code.

Example:

Total Line of Code (LOC) = 09

```
//Import header file
#include <iostream>
int main () {
    int num = 10;
    //Logic of even number
    if (num % 2 == 0)
    {
        cout<<"It is even number";
    }
    return 0;
}
```

Size Metrics : LOC

Advantages:

1. Most used metric in cost estimation.
2. It is very easy in estimating the efforts.

Disadvantages:

1. It cannot measure the size of the specification.
2. Bad software design may cause an excessive line of code
3. It is language dependent.
4. Users cannot easily understand it

Project	LOC	Cost \$K	Efforts (Persons/ Month)	Documenta tion	Errors
A	10000	110	18	365	39
B	12000	115	20	370	45
C	15400	130	25	400	32

LOC Table

Example: LOC Approach

Average productivity for systems of this type = 620 LOC/pm.

Burdened labor rate =\$8000 per month, the cost per line of code is approximately \$13.

Based on the LOC estimate and the historical productivity data, the total estimated project cost is **\$431,000 and the estimated effort is 54 person-months.**

Function	Estimated LOC
User interface and control facilities (UICF)	2,300
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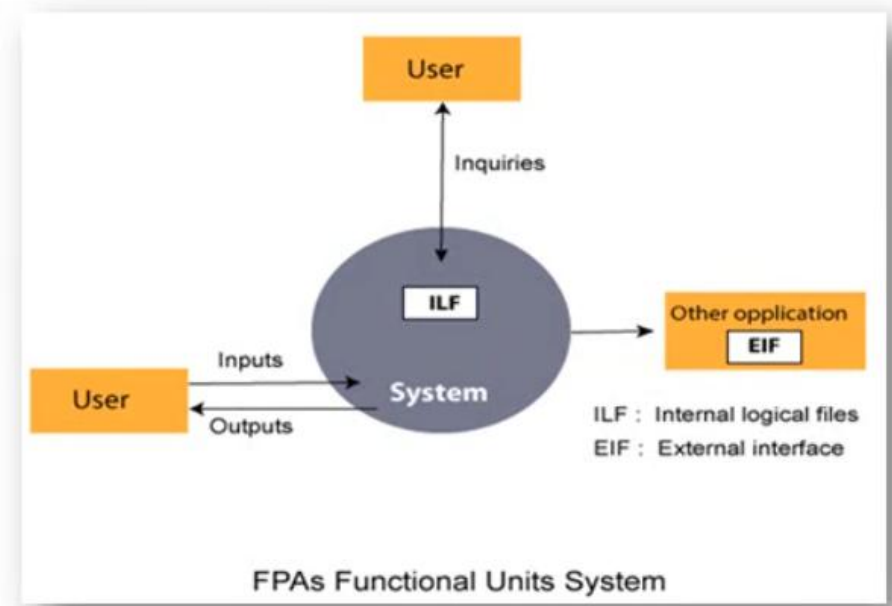
Size Metrics : FP

2. FP (Functional Point):

- Function-oriented software metrics measure functionality, what the system performs, is the measure of the system size.
- It find out by counting the number and types of functions used in the applications

FP Attributes

Measurements Parameters	Examples
1.Number of External Inputs(EI)	Input screen and tables
2. Number of External Output (EO)	Output screens and reports
3. Number of external inquiries (EQ)	Prompts and interrupts.
4. Number of internal files (ILF)	Databases and directories
5. Number of external interfaces (EIF)	Shared databases and shared routines.



Function Point (FP) based approach

- FP estimates, decomposition works differently.
- Rather than focusing on function, each of the information domain characteristics—inputs, outputs, data files, inquiries, and external interfaces—as well as the 14 complexity adjustment values
- Using historical data, the FP metric can be used to
 - Estimate the cost or effort required to design, code, and test the software;
 - Predict the number of errors that will be encountered during testing;
 - Forecast the number of components and/or the number of projected source lines in the implemented system.
- F_i - Value Adjustment Factor

$$FP = \text{count total} * [0.65 + 0.01(\sum(F_i))]$$

Value Adjustment Factors

- The F_i ($i = 1$ to 14) are *value adjustment factors* (VAF) based on responses to the following questions:
 - Does the system require reliable backup and recovery?
 - Are specialized data communications required to transfer information to or from the application?
 - Are there distributed processing functions?
 - Is performance critical?
 - Will the system run in an existing, heavily utilized operational environment?
 - Does the system require online data entry?
 - Does the online data entry require the input transaction to be built over multiple screens or operations?
 - Are the ILFs updated online?

Value Adjustment Factors

- Are the ILFs updated online?
- Are the inputs, outputs, files, or inquiries complex?
- Is the internal processing complex?
- Is the code designed to be reusable?
- Are conversion and installation included in the design?
- Is the system designed for multiple installations in different organizations?
- Is the application designed to facilitate change and ease of use by the user?

Size Metrics : FP

Advantages:

1. Need detail specification.
2. Not restricted to code.
3. Language Interdependent.

Disadvantages:

1. Ignore quality issues.
2. Subjective counting depend on estimation.

	A	B	C	D	E	F	G
	Info Domain	Optimistic	Likely	Pessim.	Est Count	Weight	FP count
1							
2	# of inputs	22	26	30	26	4	104
3	# of outputs	16	18	20	18	5	90
4	# of inquiries	16	21	26	21	4	84
5	# of files	4	5	6	5	10	50
6	# of external inter	1	2	3	2	7	14
7	UFC: Unadjusted Function Count						342
8			Complexity adjustment factor				1.17
9						FP	400

Example 1 – COMPUTING FP

Information Domain Value	Count		Weighting factor			
			Simple	Average	Complex	
External Inputs (EIs)		×	3	4	6	=
External Outputs (EOs)		×	4	5	7	=
External Inquiries (EQs)		×	3	4	6	=
Internal Logical Files (ILFs)		×	7	10	15	=
External Interface Files (EIFs)		×	5	7	10	=
Count total	→					

FP COMPUTATION – SAFE HOME System

Information Domain Value	Count		Weighting factor			
			Simple	Average	Complex	
External Inputs (EIs)	3	×	3	4	6	= 9
External Outputs (EOs)	2	×	4	5	7	= 8
External Inquiries (EQs)	2	×	3	4	6	= 6
Internal Logical Files (ILFs)	1	×	7	10	15	= 7
External Interface Files (EIFs)	4	×	5	7	10	= 20
Count total						50

$$FP = \text{count total} * [0.65 + 0.01(\sum(Fi))]$$

$$FP = 50 * [0.65 + (0.01 * 46)] = 56$$

Example 2 - Function Point

Problem Statement

Continuing the estimation of the same CAD software package, the team now wants to size the system using the **Function-Point (FP) method** instead of (or alongside) LOC, since FP is based on what the system *does* for the user rather than how many lines of code it takes to build it.

For each of the five information-domain values, the team gathers **optimistic, likely, and pessimistic** counts (the same three-point idea used for LOC), then applies a standard **complexity weight** to each:

Information domain value	Opt.	Likely	Pess.	Weight
Number of external inputs	20	24	30	4
Number of external outputs	12	15	22	5
Number of external inquiries	16	22	28	5
Number of internal logical files	4	4	5	10
Number of external interface files	2	2	3	7



The organization's average productivity is **6.5 FP/person-month**, and its average burdened labor rate is **\$8,000/person-month**. Fourteen complexity adjustment factors (installation ease, performance, reusability, etc.) have been rated, giving a total factor sum $\Sigma F_i = 52$ (on a 0–5 scale each).

Find: the estimated FP count, and from it, the project's effort and cost.

Solution

$$FP_{\text{estimated}} = \text{count total} \times [0.65 + 0.01 \times \Sigma(F_i)]$$

Where

- **UFP** = Unadjusted Function Points (obtained from counting EI, EO, EQ, ILF, EIF)
- **0.65** = Fixed constant
- **0.01** = Fixed constant
- ΣF_i = Sum of the **14 General System Characteristics (GSCs)**, each rated from 0 to 5

Solution

Step 1 — Estimate the count for each information domain value (three-point formula)

$$S = \frac{S_{opt} + 4S_m + S_{pess}}{6}$$

Example — external inputs:

$$S = \frac{20 + 4(24) + 30}{6} = \frac{20 + 96 + 30}{6} = \frac{146}{6} \approx 24$$

Applying this same formula to each row gives the **estimated count** column:

Count Total

Information domain value	Opt.	Likely	Pess.	Est. count	Weight	FP count
Number of external inputs	20	24	30	24	4	97
Number of external outputs	12	15	22	16	5	78
Number of external inquiries	16	22	28	22	5	88
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Number of external interface files	2	2	3	2	7	15
<i>Count total</i>						320

Value Adjustment Factor

Factor

Backup and recovery
Data communications
Distributed processing
Performance critical
Existing operating environment
Online data entry
Input transaction over multiple screens
Master files updated online
Information domain values complex
Internal processing complex
Code designed for reuse
Conversion/installation in design
Multiple installations
Application designed for change

Value

4
2
0
4
3
4
5
3
5
5
4
3
5
5

$$\sum F_i = 52$$

Value adjustment factor

1.17



Step 3 — Compute the Value Adjustment Factor (VAF)

$$FP = \text{count total} \times \left[0.65 + 0.01 \times \sum F_i \right]$$

$$FP = 320 \times [0.65 + 0.01(52)] = 320 \times 1.17$$

Step 4 — Calculate final FP count

$$FP \approx 375$$

Step 5 — Convert FP to effort

$$\text{Effort} = \frac{FP}{\text{Productivity}} = \frac{375}{6.5} \approx 58 \text{ person-months}$$

Step 6 — Convert effort to cost

$$\text{Cost} = 58 \times \$8,000 \approx \$461,000$$



Step 7 — Cost per function point

$$\frac{\$8,000}{6.5} \approx \$1,230 / \text{FP}$$

Result: The CAD software is estimated at **375 function points**, requiring approximately **58 person-months** of effort, at a total cost of **\$461,000** ($\approx \$1,230$ per function point) — a figure that lines up closely with the 54-person-month LOC-based estimate, giving the team confidence that both estimates are in the right range.

3. Process based Estimation

- Most common technique for estimating project is to estimate on the process that will be used.
- The process is decomposed into a relatively small set of tasks and the effort required to accomplish each task is estimated.
- Once problem functions & process activities are decided, the planner estimates the effort (e.g., person-months) that will be required to accomplish each software process activity for each software function.
- Average labor rates (cost/unit effort) are then applied to the effort estimated for each process activity.
- It is very likely the labor rate will vary for each task.
- Senior staff heavily involved in early activities are generally more expensive than junior

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